

Department of Computer Science & Information Technology



PROJECT TITLE

MOODLIFT, MENTAL HEALTH & MOOD TRACKING APPLICATION

Session: BS-CS 2022–2026

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Abstract

This project shows the development of a MoodLift: “Mental Health Wellness, Stress Relief, and Mood tracking application designed to help users to monitor and track their mood daily to improve mental health. It also provides well maintained community to discuss their condition or to take inspiration from others recovery experience. In today’s era, many people struggle to manage stress, anxiety, sleep issues, depression and changing emotions. Moodlift discover this need and provide users an application to record their regular moods, track emotions, and gets meaningful recovery ways to get rid of their mental issues.

This application allows user to answer mood related questions daily, share personal experiences by posting stories by keeping their identity hidden, and engage with a very supportive community through likes, and comments. By checking user’s activity and mood entries, application generates insights and analytics that helps user to understand their emotion and start improving by work on it.

Moodlift is developed using MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. Its features are responsive and Interface is user-friendly that displays an effective interface with efficiently working. The main objectives of this app is also to promote mental wellness, encouragement about self-reflection, and provide users a digital tool for tracking and monitoring understanding as well.

Acknowledgement

I want to say thank you much to my supervisor Dr. Muniba Saleem. She helped me a lot, with this project. I really appreciate her guidance and support. She always gave me feedback and motivated me to do my best. This really helped me to finish the project.

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Abbreviations

Abbreviation	Full Form
MERN	MongoDB, Express.js, React Native, Node.js
API	Application Programming Interface
Expo	Expo Framework for React Native
UI-UX	User Interface - User Experience
JWT	JSON Web Token
REST	Representational State Transfer
MongoDB	Mongo Database
SRS	Software Requirements Specification
UAT	User Acceptance Testing
ERD	Entity Relationship Diagram
CRUD	Create, Read, Update, Delete
JSON	JavaScript Object Notation
WBS	Work Breakdown Structure

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Chapter 1: Final Project Proposal

1.1 Introduction:

Mental health remains a critical factor in everyone's life and at present many are finding it hard to manage daily emotions given their occupied and busy life schedules with study and work and other personal responsibilities. By keeping an eye on their day to day mood they learn patterns about themselves and can identify triggers and form sustainable habits. MoodLift a mobile application designed to improve the well-being of every one of its user to monitor their daily moods & emotion throughout there day.

MoodLift is a mobile application developed to support mental wellness by allowing users to monitor their daily moods and emotions. This application is built using React Native (Expo) for the mobile frontend, Node.js and Express.js for the backend, and MongoDB as the database. This technology stack allows the application to provide a responsive, secure, and smooth user experience on their devices.

The application allows users to record their daily moods, answer wellness related questions, share personal stories, and interact with the community through likes and comments that helps to find a best solution or way to get rid of mental issues like stress, depression, sleep issues etc. Based on user's mood history and activities, MoodLift generates mood insights and activity based analytics that help users understand their emotional patterns and encourage positive mental health practices.

The main goal of MoodLift is to provide an easy to use digital platform that promotes self-awareness, emotional relaxation, and community support while maintaining both user's privacy and data security.

1.2 Problem Statement

As we all know that, mental health issues are becoming increasingly very common, rather than this many people do not have access to use simple tools that help them monitor and understand their emotional well-being. Everyone often experience their mood changes caused by stress, academic workload, work pressure, or personal problems, but these emotions are rarely recorded or analyzed. Without proper tracking, users may fail to

recognize emotional patterns that could negatively affect their mental health.

Regardless of the existence of multiple mood tracking apps, the vast majority of them only record mood and fail to offer social connectivity, emotional analytics, and personalized insights. Some applications are difficult to use, while others require paid subscriptions to access useful features. And just because of this paid feature, people leave that app to track moods and get rid of mental health issues.

To overcome these challenges, the MoodLift app offers a complete mobile application platform where users can record daily moods, answer wellness-related questions, share personal experiences, ask for recovery ways from the community, interact with others through likes and comments, and receive mood insights based on their activities.

The application encourages users to become more aware of their emotional health and supports them in developing healthier daily habits through continuous mood monitoring.

1.3 Project Goals and Objectives

The main goal of the MoodLift project is to develop a user-friendly mobile application that helps everyone to monitor their emotions, identify mood patterns, engage with very cooperative community, and encourage positive mental health practices.

The specific goals of the project are:

- To develop a secure mental wellness platform using the MERN Stack.
- To allow users to record and manage their daily mood entries.
- To deliver a daily wellness question for self-reflection.
- To enable users to share personal stories with the community.
- To allow community interaction through likes, comments and sharing stories.
- To generate activity-based analytics and mood insights from the data of recorded mood.
- To provide a responsive and attractive user interface that works on both desktop and mobile devices.
- To maintain user privacy by implementing secure authentication and authorization.
- To foster a sense of consistency in mood and habits to encourage healthier habits.

1.4 Tools and Technologies Used

MoodLift is developed using modern technologies that ensure high performance, scalability, and ease of maintenance. The following tools and technologies were used during the development of the project:

Table 1.1: Tools and technologies used:

Tools	Purpose
React.js	Used to develop frontend user interface.
Node.js	Provides the runtime environment for backend development.
Express.js	Handle server side logics and Restful APIs.
MongoDB	Stores user information, mood entries, stories, comments and other application data.
Mongoose	Connect Node.js with MongoDB and manages database schemas.
JWT (JSON Web Token)	Provides secure user authentication and authorization.
Bcrypt	Encrypts user passwords before storing them in the database.
Postman	Used to test backend APIs before frontend integration.
Visual Studio Code	Primary code editor used during development.

These technologies work together to create a secure, responsive, and scalable application which is capable of managing user data efficiently while providing a smooth user experience.

Chapter 2: Feasibility Study and Project Planning

2.1 Introduction

A feasibility study is carried out before developing a software project to determine whether the proposed system can be successfully implemented. It helps check the practicality of the project in terms of technology, operations, resources, time, and overall project planning.

Conducting a feasibility study reduces development risks and ensures that the project objectives can be achieved within the available constraints.

For the MoodLift mobile application, the feasibility study was performed to analyze whether the application could be developed using the available technologies, skills, and resources. As MoodLift is developed using the MERN stack with React Native (Expo), it is technically suitable for cross-platform mobile application development. The study also examines how the application will operate in real world scenarios and presents the project planning through a Work Breakdown Structure (WBS), which divides the project into manageable tasks.

2.2 Technical Feasibility

Technical feasibility concerns itself with whether the necessary technologies, tools, and technical resources are available for the successful implementation of the proposed system. The underlying technology Stack for Moodlift, we develop a modern, cutting edge app based on proven modern and widely used Technologies. Frontend with React.js, Native and expo. We create a cross platform app with single codebase that runs on both iOS and android platforms. Backend with Node.js and Expres.js. We develop Restful APIs to communicate between frontend and backend, and we use NoSQL MongoDB to Store all application data (Users, Moods, stories, comments and). Our development team has enough knowledge of JavaScript, React Native, Nodejs, Express.js and MongoDB to execute this project. The tools for development are Visual Studio code, Post man for APIs testing, and Expogo for testing app on android mobile phone. Our hardware requirements consist of nothing more than a laptop with at least 8 GB RAM, an active internet connection, and an android phone to test the app. Thus we can safely conclude that we have met the criteria for technical feasibility.

Technical Feasibility Shown below:

Table 2.1: Technical Feasibility

Component	Technology Used
Frontend	React Native (Expo)
Backend	Node.js and Express.js
Database	MongoDB
Programming Language	JavaScript
Authentication	JWT & Bcrypt
API Testing	Postman
Development Environment	Visual Studio Code

2.3 Operational Feasibility

Operational feasibility focuses on how the suggested solution will be performed and used within the given context and if it meets user's needs. The interface provided is sleek and visually appealing as well as intuitive and user friendly, ensuring non-technical user can use the application and operate the functionalities without much learning. Users can be registered, logged into the application, log their mood, take wellness quiz, write stories, like and comment on stories. Users will be gaining insights to their moods and activity based analysis helping them be aware of emotional states on longer term. Responsiveness and compatibility of user interface over Mobile applications gives accessibility across the board and at any point of time. Application backend ensures that user data will be properly guarded with authentication and authorization system implemented for data security and to grant

secure access. User's password should be encrypted or hashed when they are to be stored, with limited and approved users having access to personal data and application features.

2.4 Work Breakdown Structure (WBS)

A Work Breakdown Structure (WBS) is a project management tool which helps to “Deconstructs a work into smaller task” which ensures that the project management activities are carried out efficiently and effectively. The Work Breakdown Structure provides framework through which project activities can be organized, and assignments delegated.

Moreover, it can provide a clear mechanism through which work completion stages could be supervised during the life cycle. Work Breakdown Structure for the MoodLift project including the several phases, including requirement gathering, system design, frontend development, backend development, database implementation, testing, and documentation.

By structuring the work into small task the whole process of development of the MoodLift can be supervised properly and the quality as well as time management is assured to improve.

Table 2.2: Work Breakdown Structure

Phase	Activities
Requirement Analysis	Gather Requirements, define project scope, identify user needs
Project Planning	Create proposal, feasibility study, and project schedule.
System Design	Design UI, architecture, ER Diagram, user journey, sequence diagram.
Frontend Development	Develop authentication, dashboard, mood tracking, stories, and community screens.
Backend Development	Create APIs, authentication, database integration, controllers
Database Development	Design MongoDB collections, and relationships

Phase	Activities
Testing	Unit testing, integration testing, bug fixing, and system testing.
Documentation	Prepare report, diagrams, screenshots, user manual

Work breakdown Structure Diagram:

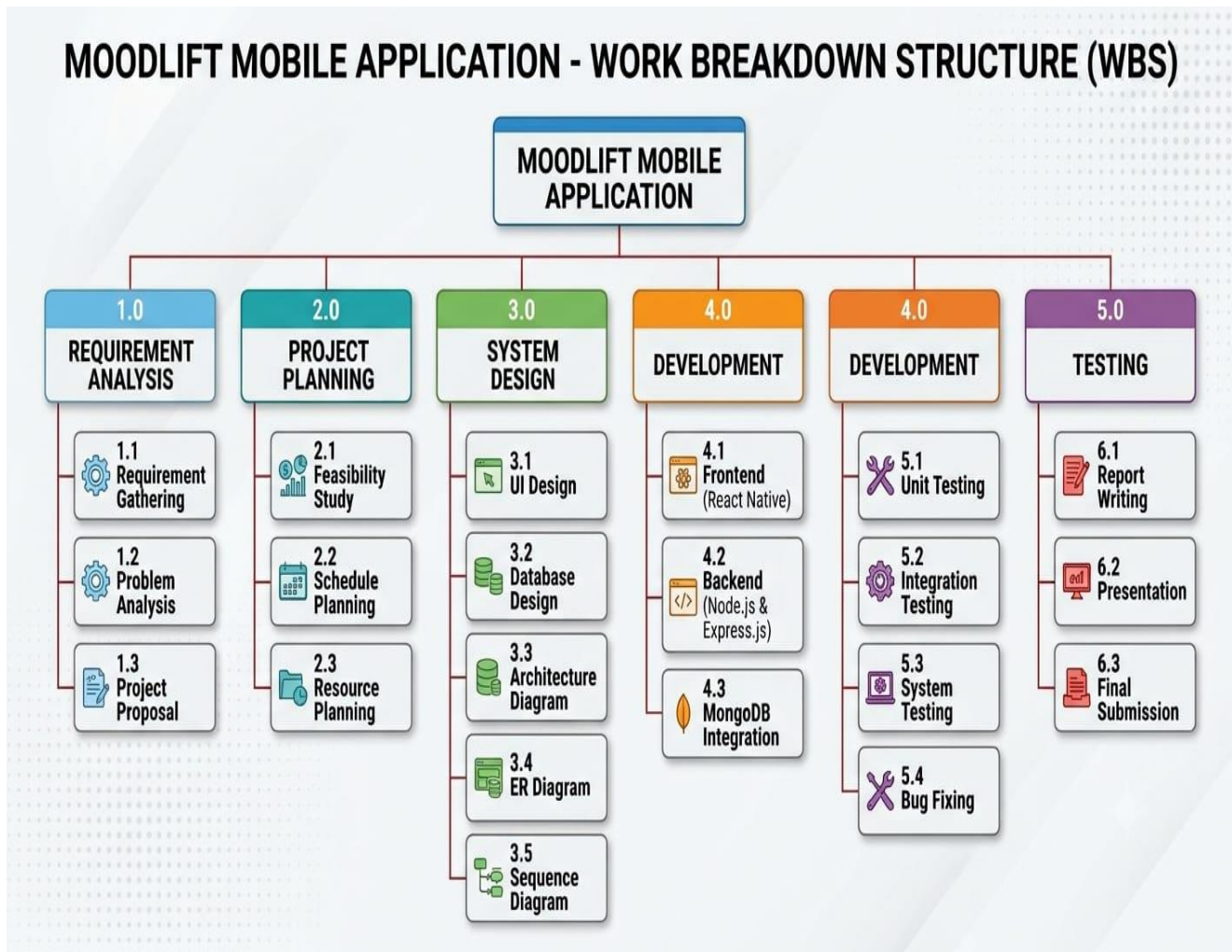


Figure 2.1: Work Breakdown Structure Diagram

Chapter 3: Software Requirements Specification (SRS)

3.1 Introduction:

The Software Requirements Specification (SRS) refers to the functional and non-functional requirements of MoodLift application. It focuses as a blueprint for the development process by clearly describing the system's features, user requirements, and overall design of the application. The Software Requirements Specification (SRS) helps developers, supervisors, and stakeholders to understand that how the application is expected to perform functions.

As MoodLift is a mobile application developed using JS Stack including React Native (Expo), Node.js, Express.js, and MongoDB. The application is mainly designed to help users to monitor their emotions by recording daily moods, answering wellness-related queries, sharing personal experiences via stories, and interacting with other users through likes and comments or with joining the community. It also generates mood insights and activity analytics that allow users to identify emotional patterns daily and helps to work better on it.

The SRS chapter explains the software requirements, system models, design principles, frontend and backend architecture, database design, and overall system architecture of MoodLift Application. The information in this chapter provides a complete understanding about how the application is designed and how its different components work together to achieve the project objectives or application's goal.

3.2 Requirement Analysis:

Requirement analysis includes mainly application's requirements such as "Functional Requirements, Non-Functional Requirements, and User Requirements"

3.2.1 Functional Requirements:

Functional requirements show the main operations and services that the application must perform. These requirements define how users interact with the system and what features the application provides to the user.

Table 3.1: Functional Requirements of MoodLift

Requirement ID	Functional Requirement	Description
FR-01	User Registration	User can easily create new account
FR-02	User Login	Registered users can easily log-in to their accounts
FR-03	Mood Tracking	Users can record their daily mood
FR-04	Daily Wellness Que	User can answer, mood related questions
FR-05	Story Sharing	User can publish personal experience
FR-06	Community Interaction	User may take advantage of other's recovery story
FR-07	Moon Analytics	System analyze mood history and generates insights
FR-08	User Profile	User can easily update their information
FR-09	Secure Logout	User can safely log-out their account

Description of Functional Requirements

a. Registration:

MoodLift application allows people to create an account by submitting required information. The system checks the entered information whether they are correct or not before creating the account.

b. Authentication:

Registered users can easily log-in to the application. The authentication of application is checked by JSON Web Tokens (JWT). It ensures that only legitimate users can access application's features.

c. Mood Tracking:

Users can record their current mood daily. They may also include additional notes describing their feelings. Application stores this information for future analysis.

d. Wellness Questions:

MoodLift provides daily questions that encourage users to observe their emotions and mental health. User responses are securely stored.

e. Story Sharing:

Users can share personal experiences with the MoodLift community by posting stories. Stories help encourage positive discussions and emotional support between users.

f. Community Interaction:

Users can engage with community posts by liking stories and sharing or posting comments. This feature increases user engagement and creates a supportive environment or community.

g. Mood Analytics:

Based on recorded moods of the user and user activities, the application generates insights that help users to understand emotional trends and patterns.

h. Profile Management:

Registered users can view, check, and update their personal information whenever needed or whenever they want to update or edit.

i. Logout:

Registered users can easily end their session, which prevent unauthorized access to their account.

3.2.2 Non-Functional Requirements

Non-functional Requirements tell how well things operate, versus the capabilities that have already been discovered. They are often called the quality requirements.

Table 3.2: Non-Functional Requirements

Category	Non-Functional Requirements
Performance	Make sure that the application perform is very quick and doesn't lose any uptime.
Security	Use bcrypt to hash the passwords and JWT to authenticate the users.
Reliability	The application should provide stable and smooth performance with minimal downtime.
Scalability	The backend should support an increasing number of users and mood records equally.
Availability	Users should be able to access the application whenever an internet connection is available on their devices.
Maintainability	The project should follow a modular structure for easier updates and maintenance of the application.
Usability	The user interface should be simple, attractive, and easy to understand even for non-tech people.

Explanation:

At MoodLift, our focus on a positive experience for you comes from prioritizing usability, security, and performance. Hence, responsive mobile interface, authentication in a flash, API responses, safe database usage.

3.2.3 User Requirements

User requirements describe the needs and expectations of the people who will use the application. Some user requirements are:

- Users will create an account quickly.
- Users will login securely with credentials.
- Users will record daily moods easily.
- Users will answer daily health questions.
- Users will submit personal story to community.
- Users will like and comment on posts.
- Users will view their daily moods and statistics.
- Users will update profile whenever they need.
- Users will use app without any prior technical skill.

The app has been kept simple to be used and enjoyed by users of all ages.

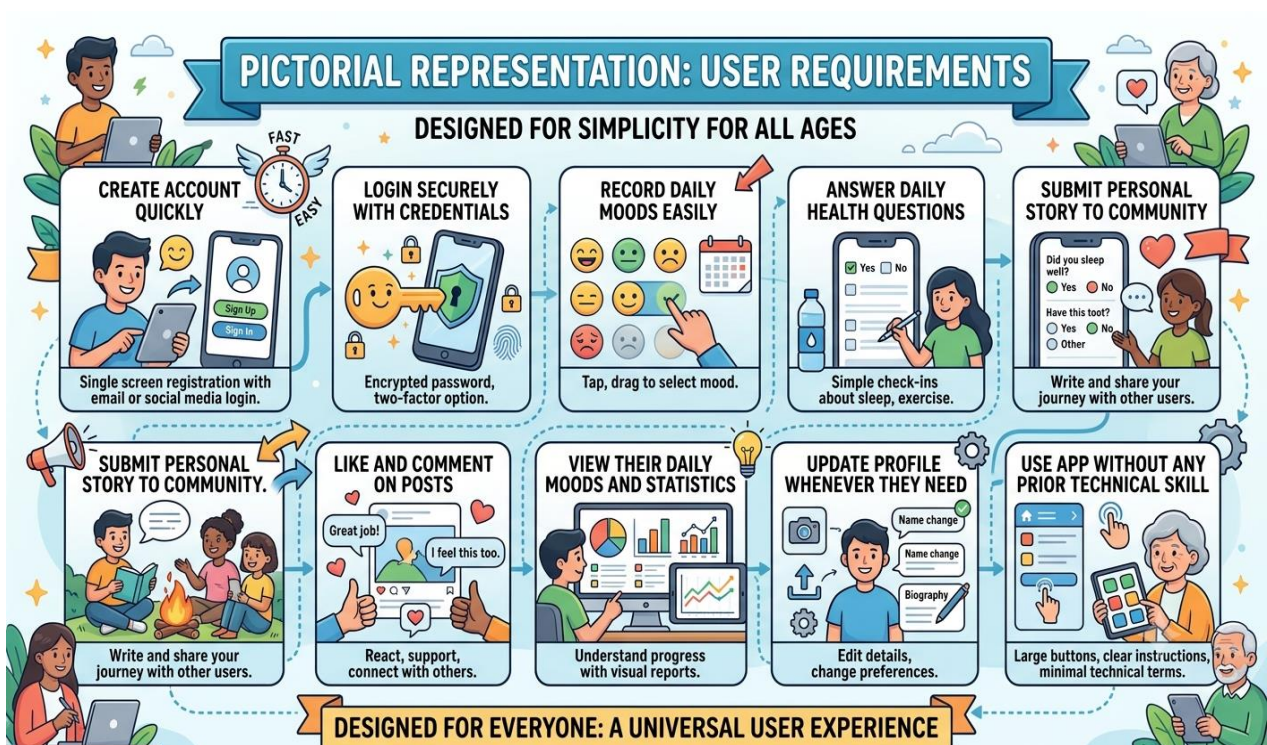


Figure 3.1: User Requirements

3.3 System Modeling

A System Model represents the system by providing visual and textual diagrams that show how individual system components behave with other systems or people. They give an insight into which the data and messages between individual applications is flowing and thus show which operations may be applied to a certain set of entities. As developers, System Model helps define what the system should be and may provide insights into system design issues by providing means to describe, explain, explore or improve systems of concern.

In the MoodLift mobile app, System modeling contains Context Diagram, Use case Diagram and Description, and Sequence Diagram.

All of them define the information flowing among the system and the users and illustrates all the important functions provided by the system.

3.3.1 Context Diagram

The central external stakeholder of MoodLift app is the registered user. He can interact with the app via Registering, Login, Create log of mood, Respond to daily question, share a story, like an article/post, comment on user's story, read an insightful view of moods. These inputs of user, the app shall handle them, save to database and response back to the user accordingly.

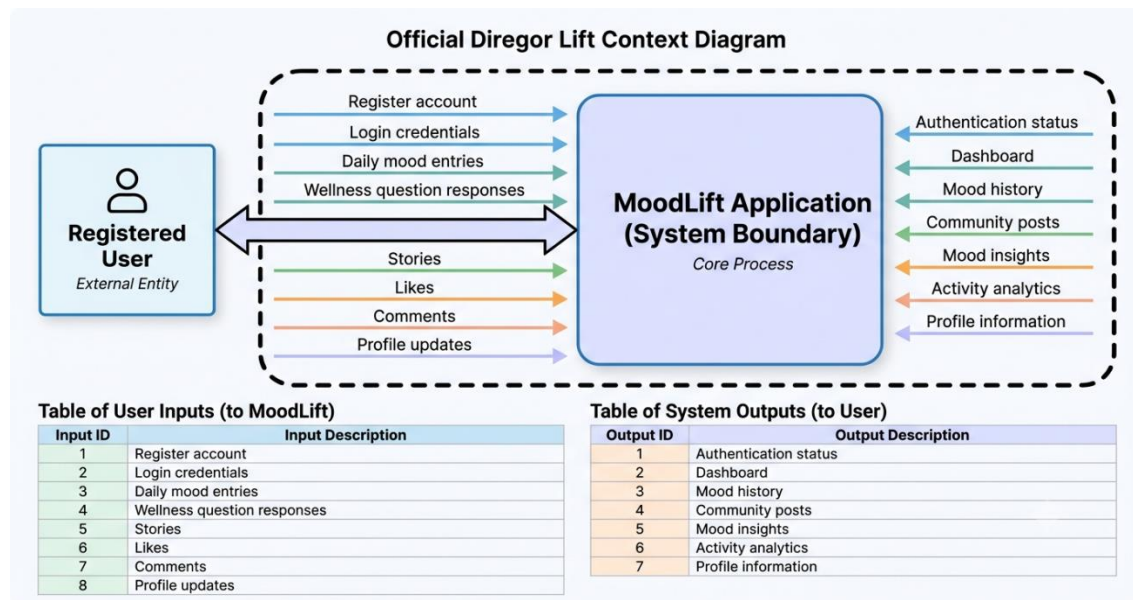


Figure 3.2: Context Diagram

3.3.2 Use Case Diagram

The Use Case Diagram explains the relationship between the user and various functionality of MoodLift Application. The main focus is on explaining various actions a user could have to do after user interacting with the system or application. In this the main actor will be the 'User' of MoodLift Application. The user can after the authentication checking, he can perform different activities. The major use cases identified for MoodLift Application are:

Register account, Login, View dashboard, Record daily Mood, Answer daily Questions, View moods History, Sharing story, Liking story, Commenting on story, View Mood's insights, Update Profile, And Logout.

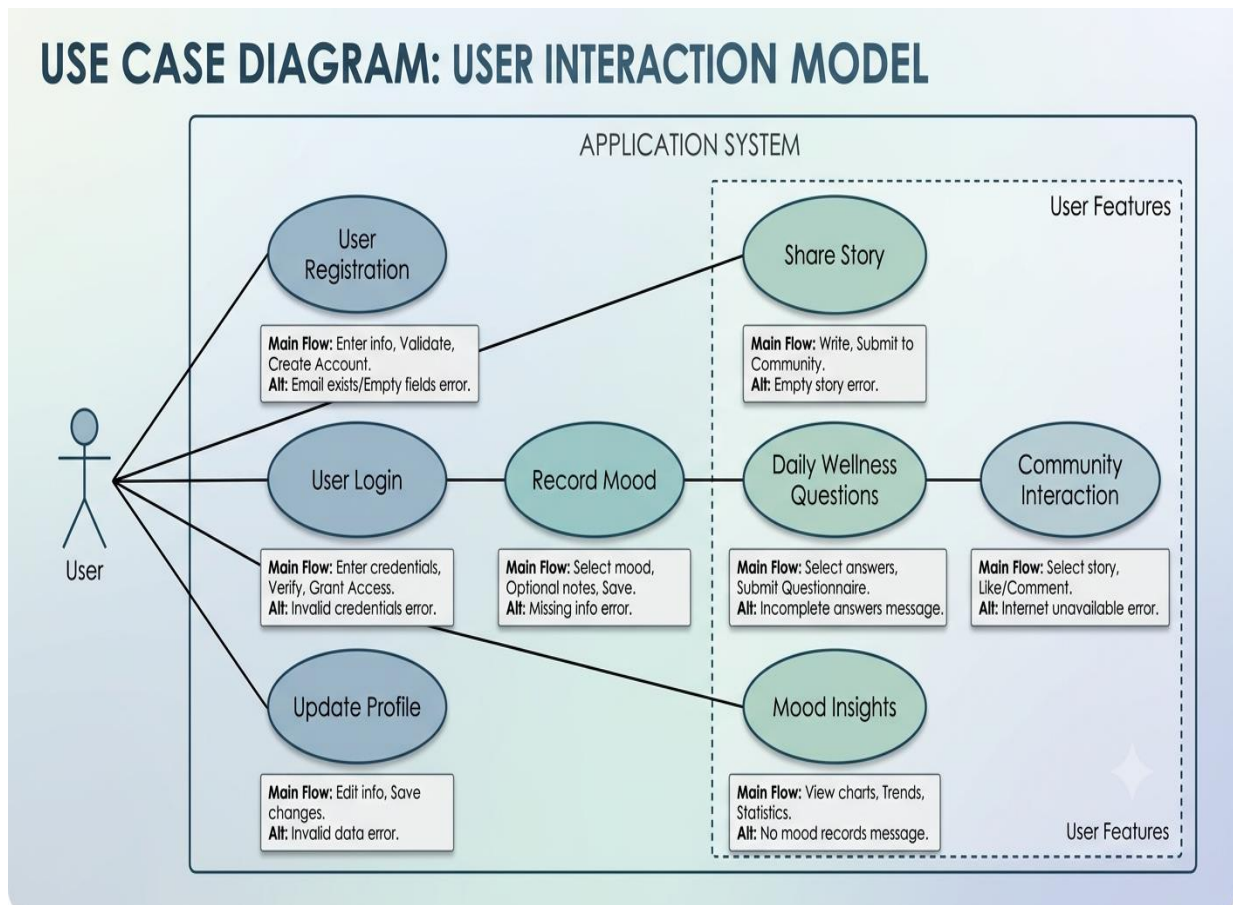


Figure 3.3: Use Case Diagram of MoodLift

3.4 Sequence Diagram

The sequence diagram in Figure 3.4 below describes the flow of requests among the user, various components of MoodLift App, Node.js server, and the MongoDB database. Let's see it sequence starts by the user:

1. The user logs in, records a mood, posts a story or checks mood-related analytics.
2. React native application send a request towards our backend (node.js, express.js).
3. Our node server receive the request validate the authenticity and process relevant logic with the backend MongoDB database and respond back towards our app and show to user results.

Main Components:

User, React Native Mobile Application, Node.js & Express Backend, MongoDB Database

Sequence Flow

1. When a user starts their device to use the application on mobile.
2. When a user enters the system using the login form to get access into the web app and for that
3. Mobile app will post on a specific HTTP method on the backend to validate user info.
4. Backend will check that the posted value in the username field for validity.
5. On successful completion; the check from MongoDB will proceed to verify the info available for that user.
6. Once, validated; then generate a JWT Authentication Token.
7. Now after login; redirected to the user Dashboard.
8. Log their mood or an activity by entering text to it or by any option available
9. Now store on database or to pull from the database information stored based on the logged activity/mood or any selection performed in the Dashboard.
10. The backend sent back the success response for the logged activity.
11. Show updated details from backend onto the user dashboard view.

Sequence Diagram of MoodLift

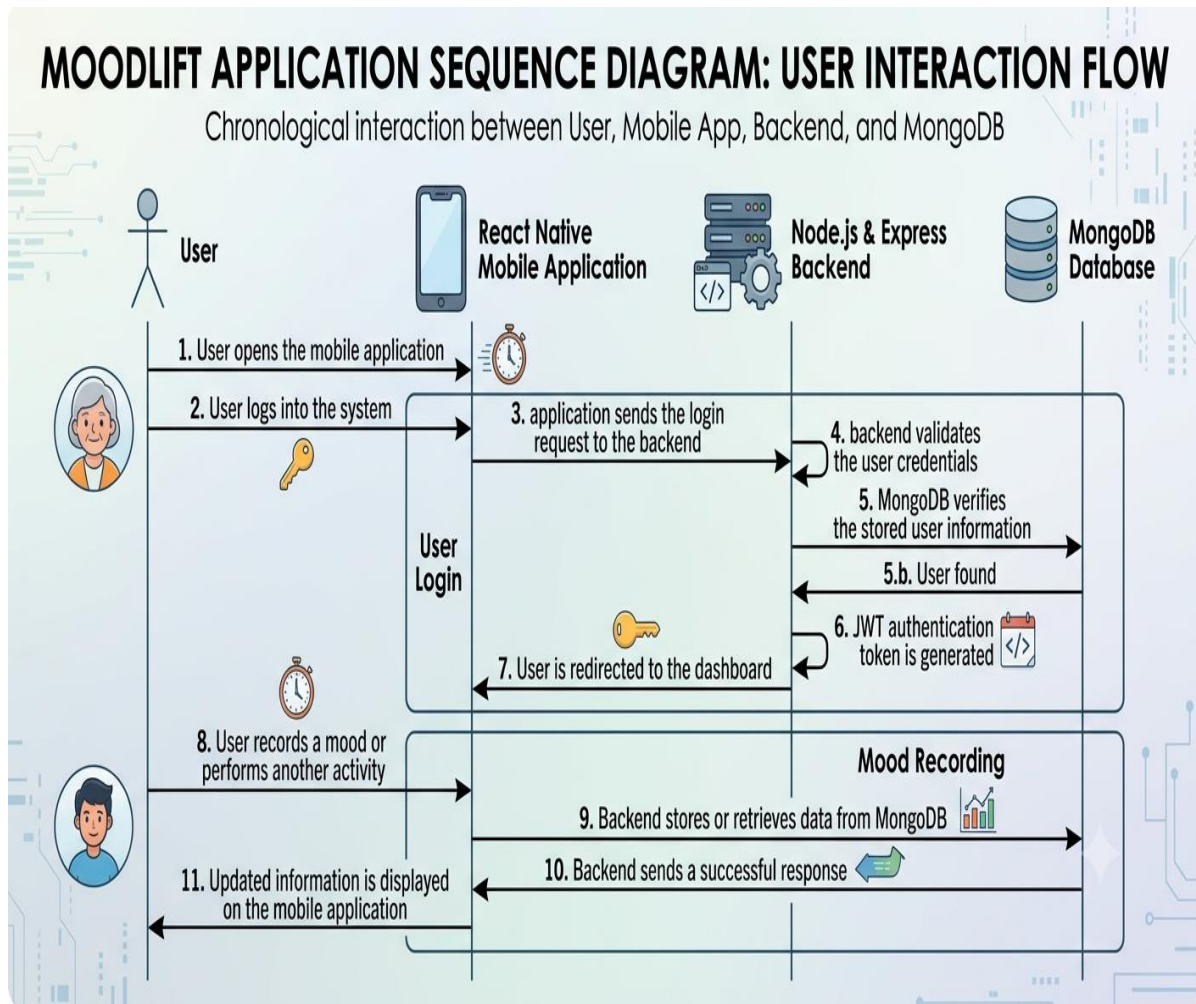


Figure 3.4: Sequence Diagram of MoodLift

3.5 Data-Flow Diagram

The flowchart shows overall functionality of the MoodLift app, showing how a user interacts with it. It describes the entire process from the moment the user opens the app to the user signing out of the app. Once the user successfully authenticates into their account, they view the dashboard where they can do such things as track their mood, answer daily wellness questions, submit story posts, view and engage with posts from the community, and view their mood insights. Each time the user takes an action, the backend processes that action and will either store that data in the MongoDB database or retrieve it from there.

Data-Flow Diagram

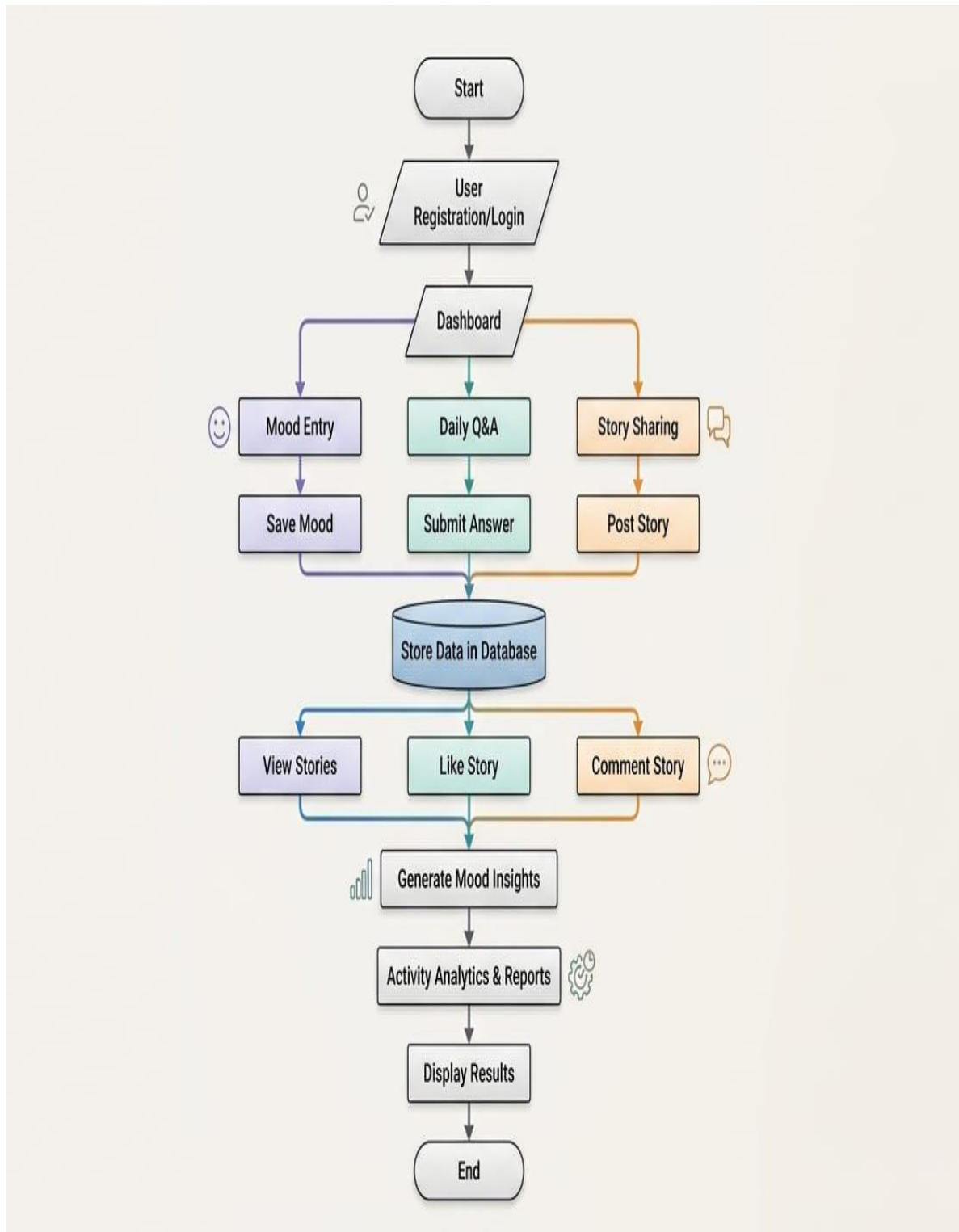


Figure 3.5: Data-Flow Diagram

3.6 Design System

A design system provides designers and developers with a common vocabulary of reusable design patterns, styles, and rules for building out consistent user interfaces across the whole application, which promotes consistent visuals like layout, colors, typography and overall component behavior across many screens.

With MoodLift the focus was to ensure that the entire interface was simple, accessible and emotionally reassuring, especially considering it's a mental wellbeing application we wanted to offer a relaxing feel with subtle colors, easy typography and easy interactions.

3.6.1 Design Principles

Design principles have been followed for the development of mood tracking and mood sharing application, MoodLift:

1. **Simplicity:** The UI/UX has been kept plain and clear in order to let the user understand and navigate with least amount of effort.
2. **Consistency:** Consistency in colors, fonts, navigation buttons, and text input components and other UI components have been taken into account.
3. **User-Centered Design:** This application has been design taking into the considerations the needs of the users and hence key feature such as mood tracking, sharing stories, and mood insights are placed prominently on the home page dashboard.
4. **Accessibility:** Appropriate readable size of text, adequate colors with proper contrasts, simple intuitive navigation all has been kept into considerations to be usable by any age group users.
5. **Responsive Design:** The UI layout of application adapts well with different mobile screens sizes and ensures good UI experience over any size.

3.6.2 Color Palette & Typography

Using and soft hues colors, the MoodLift app establishes a serene environment to help promote emotional wellness.

Color Palette used in MoodLift:

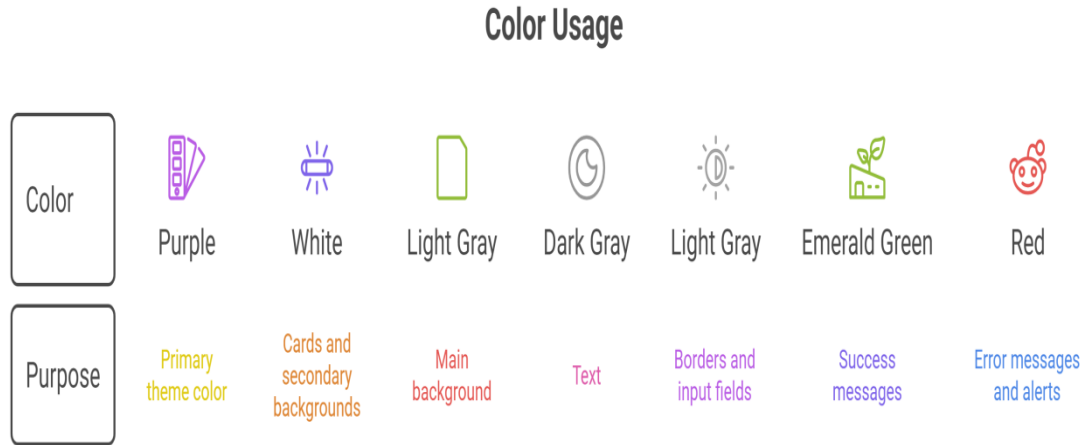


Figure 3.6: Color Usage

Typography

The typography uses an easy to read and modern font throughout the application interface. Using a contemporary and accessible font face ensures that it looks clean and works perfectly well at any size.

Which font style should be used for different text elements?

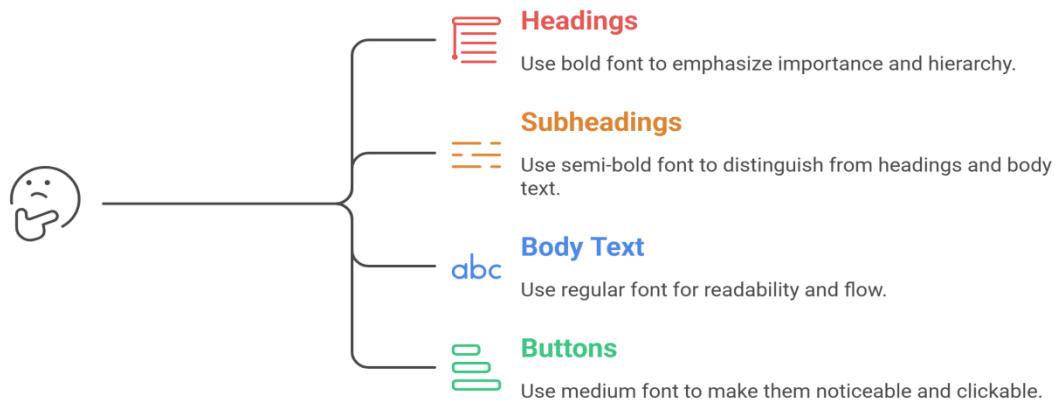


Figure 3.7: Typography used in MoodLift

Because MoodLift is a mental wellness app, it has chosen a typography style that achieves an overall clean, modern, and user-friendly presentation creating an overall

calm, comfortable reading experience. To achieve this, the application uses the two most common fonts when brainstorming for an application: Poppins and Inter. Both of these fonts are very well known as having excellent legibility and clarity for use on mobile devices.

3.6.3 Icons & Components

The MoodLift App utilizes simple but great icons to enhance users' connection with the app and enhance their overall experience.

Each element is consistently designed in accordance with style guides used on all app elements.

By using such reusable components, support will also be easier because there will be less time spent maintaining them after they are built.

Figure 4.2 shows list of components used in MoodLift

It has chosen a style that achieves an overall clean, modern, and user-friendly presentation creating an overall calm, comfortable reading experience.

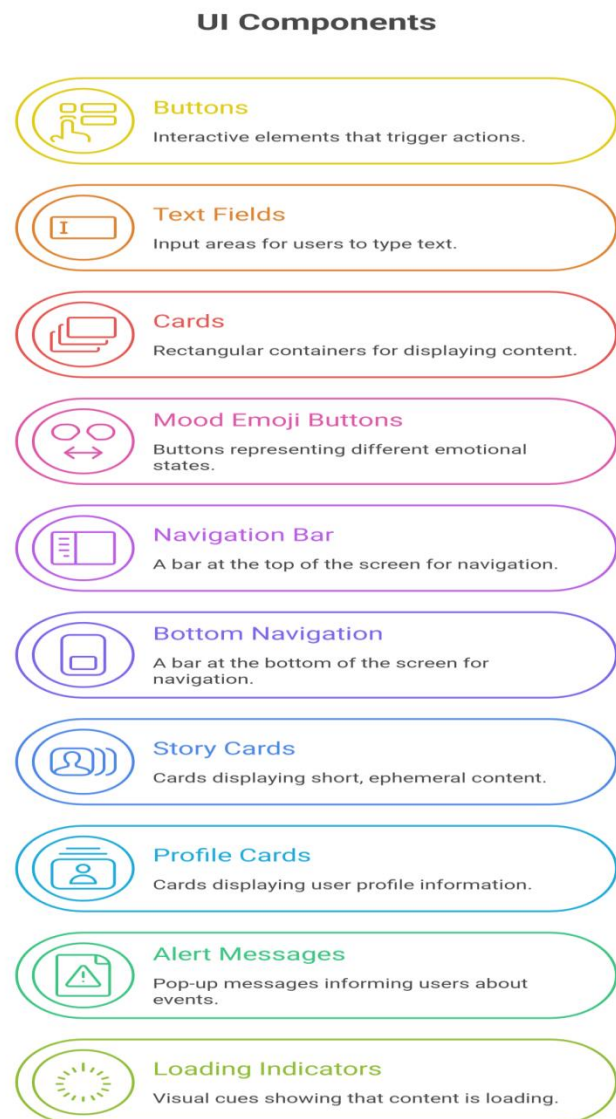


Figure 3.8: Components used in MoodLift

3.7 Frontend Design

Using React Native as the base framework for the MoodLift APP (app), and using the Expo client for the front-end of the app (user interface and experience), this enhances UI/UX design for improved user interactivity and responsiveness to accessing any function in the app; a REST API will be used to connect the back-end to the front-end of the app so both user retrieval and user storage will occur in real-time. The MoodLift APP is designed using a component driven development methodology to create a modular (i.e., reusable) code base and to create a cleaner code structure. The MoodLift user interface has minimal requirements to navigate pages from one to another; providing a smooth user experience, while providing easy access to application content.

3.7.1 Frontend Architecture of MoodLift

MoodLift's Frontend Architecture is comprised of the following screens, as illustrated below:

Log In, Sign Up, Home Dashboard, Mood Tracker, Daily Questions, Community, Story Details, Mood Insights, Profile, Settings

Additionally, the application flows following sequence:

User → React Native Interface → REST API → Backend → MongoDB

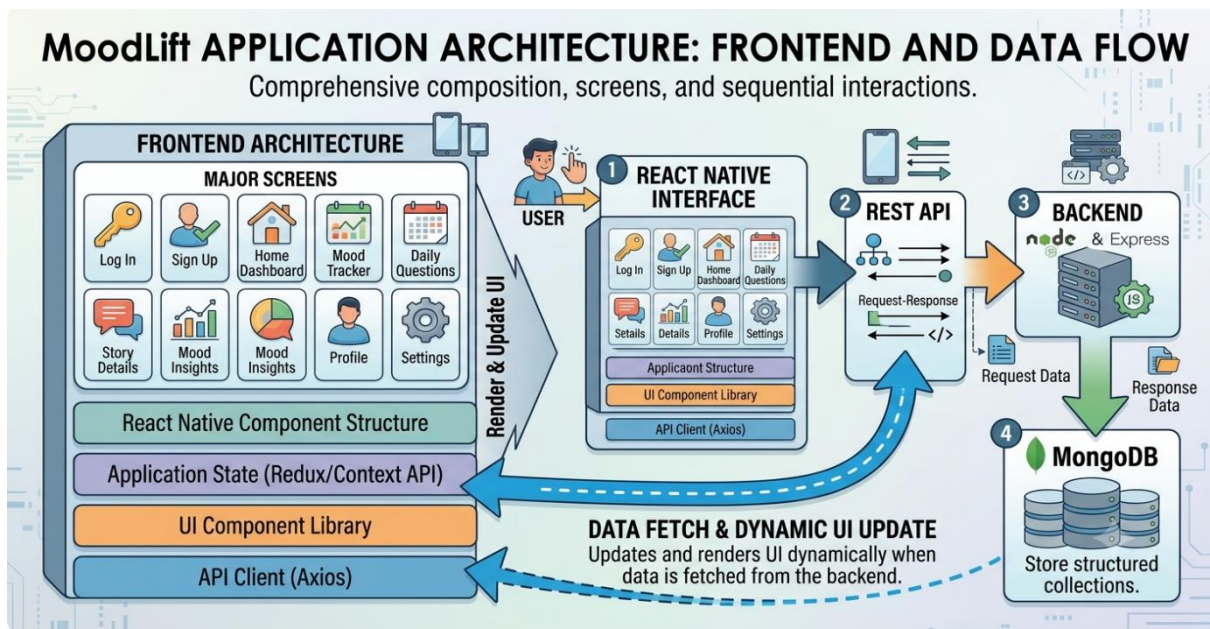


Figure 3.9: Frontend Architecture of MoodLift

3.7.2 User Interface Design of MoodLift

The UI consists of multiple screens designed for different purposes, including:

1. Log In

The Log In screen provides registered users with a safe and secure access point to the application by entering their email address and password.

Features of this screen include:

Entering Email

Entering Password

Login Button

Registration Button

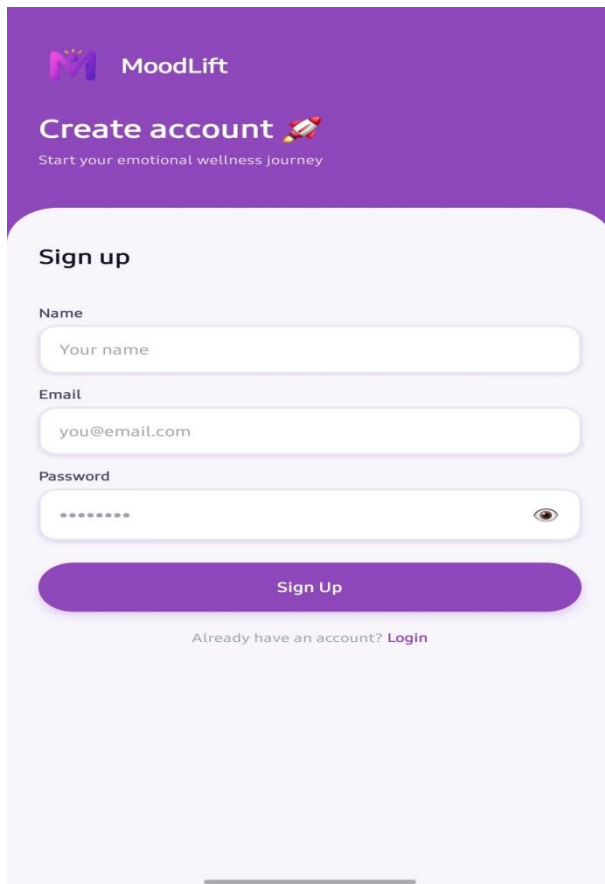
The image shows the 'Sign up' screen of the MoodLift application. The header is purple with the MoodLift logo and the text 'Create account' with a rocket icon. Below the header, the text 'Start your emotional wellness journey' is displayed. The main content area is white with a purple border. It contains a 'Sign up' title, followed by three input fields: 'Name' (placeholder: 'Your name'), 'Email' (placeholder: 'you@email.com'), and 'Password' (placeholder: '*****' with an eye icon). A purple 'Sign Up' button is at the bottom, and a link 'Already have an account? Login' is below it.

Figure 3.11: Sign-Up

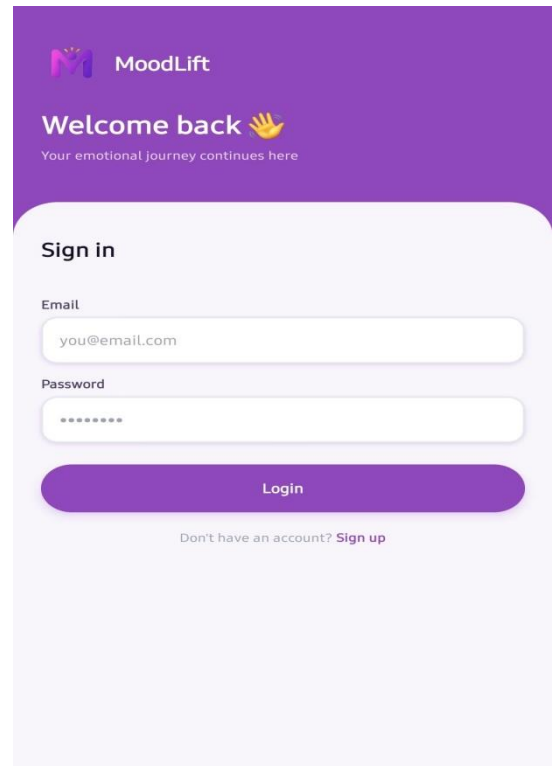
The image shows the 'Sign in' screen of the MoodLift application. The header is purple with the MoodLift logo and the text 'Welcome back' with a hand icon. Below the header, the text 'Your emotional journey continues here' is displayed. The main content area is white with a purple border. It contains a 'Sign in' title, followed by two input fields: 'Email' (placeholder: 'you@email.com') and 'Password' (placeholder: '*****'). A purple 'Login' button is at the bottom, and a link 'Don't have an account? Sign up' is below it.

Figure 3.10: Sign-In

2. Sign Up

The Sign Up screen allows for new users to create an account in the application by providing:

Features of this screen include:

Entering name

Entering email

Entering password

Sign-Up Button

Login Option

3. Home Dashboard

The home dashboard acts as a main screen after successful login and provides the user best access to:

Features of this screen include:

Mood Tracking

Mood Insights

Mood Related Questions

Community Stories

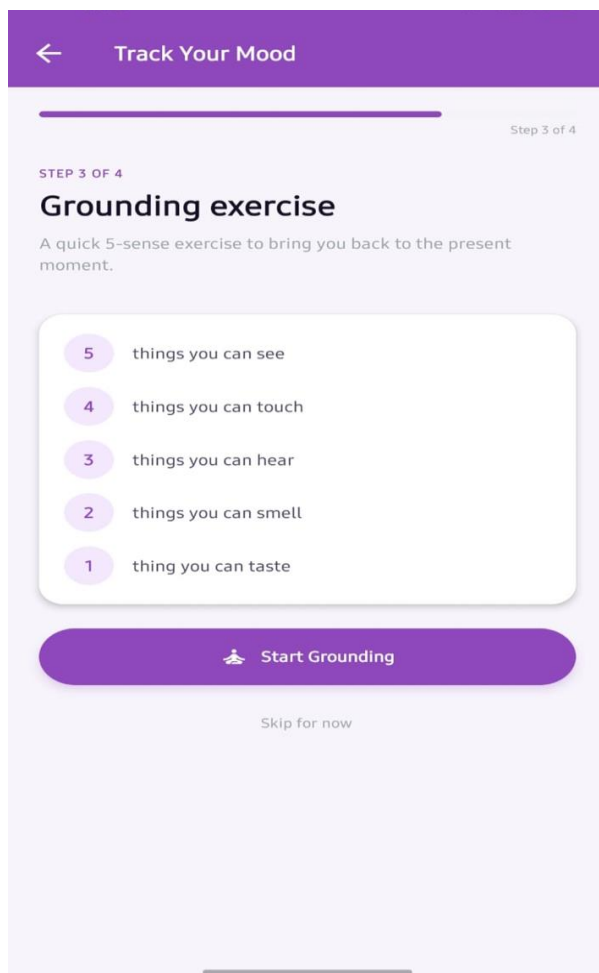


Figure 3.13: Tracking Screen

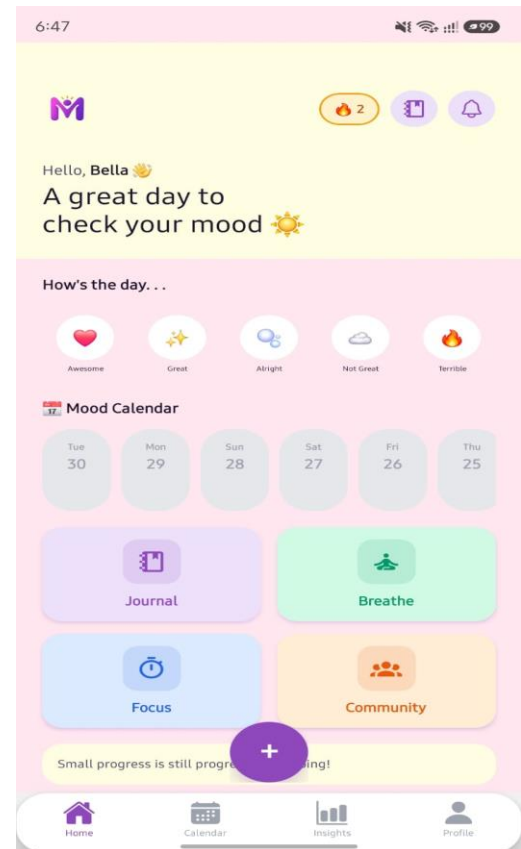


Figure 3.12: Dashboard

4. Tracking Screen

Users of the MoodLift application can track and monitor their moods with the Mood. The user's recorded mood will be saved into a MongoDB database and later analyzed to create analytics

This screen allows users to:

Select a mood, Select an emoji, Write notes about the selected mood, Save their recording

5. Community Screen

The Community Screen allows users to interact with each other in the MoodLift application. This feature fosters positive connection and emotional support through positive interactions with other users.

Users can:

Read other users' stories

Like other users' stories

Comment on other users' stories

Share their stories

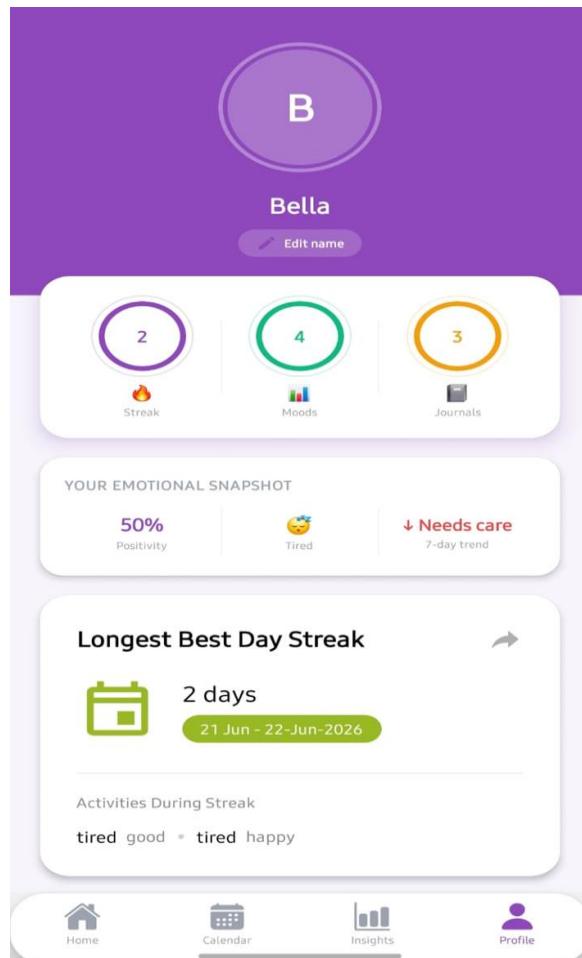


Figure 3.15: Profile Screen

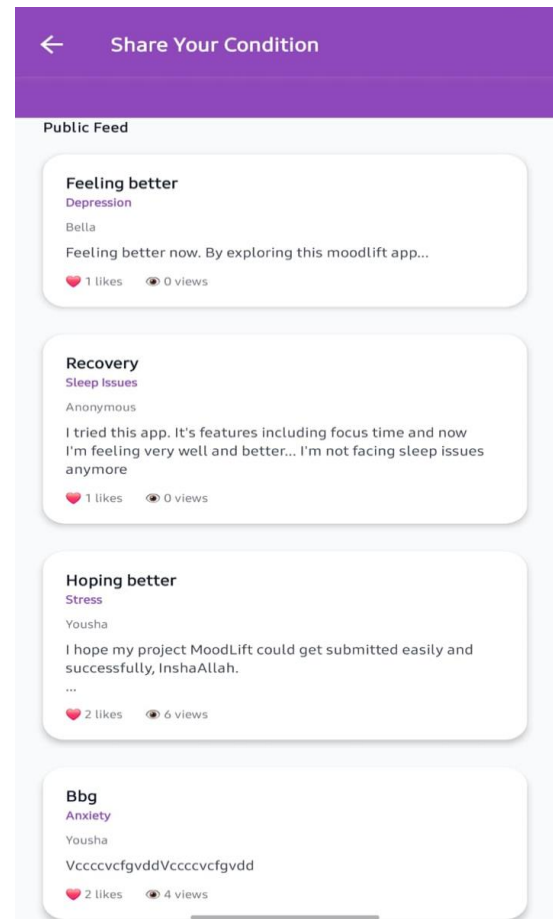


Figure 3.14: Community Screen

6. Profile Screen

The Profile Screen allows users to update their information as well as to edit their information into the existing information.

In this users can:

View their information, change their profile information, change their profile picture, Edit their name, and Logout of MoodLift

3.7.3 Community Interaction

Users can interact with one another through community interactions, which is one of the main functions of the MoodLift application.

Users can:

Share their experiences, Read community stories, Like community stories, Write comments on community stories, and Be available to offer support through positive engagement with one another. The MoodLift community provides a supportive environment for users to share and be connected to each other.

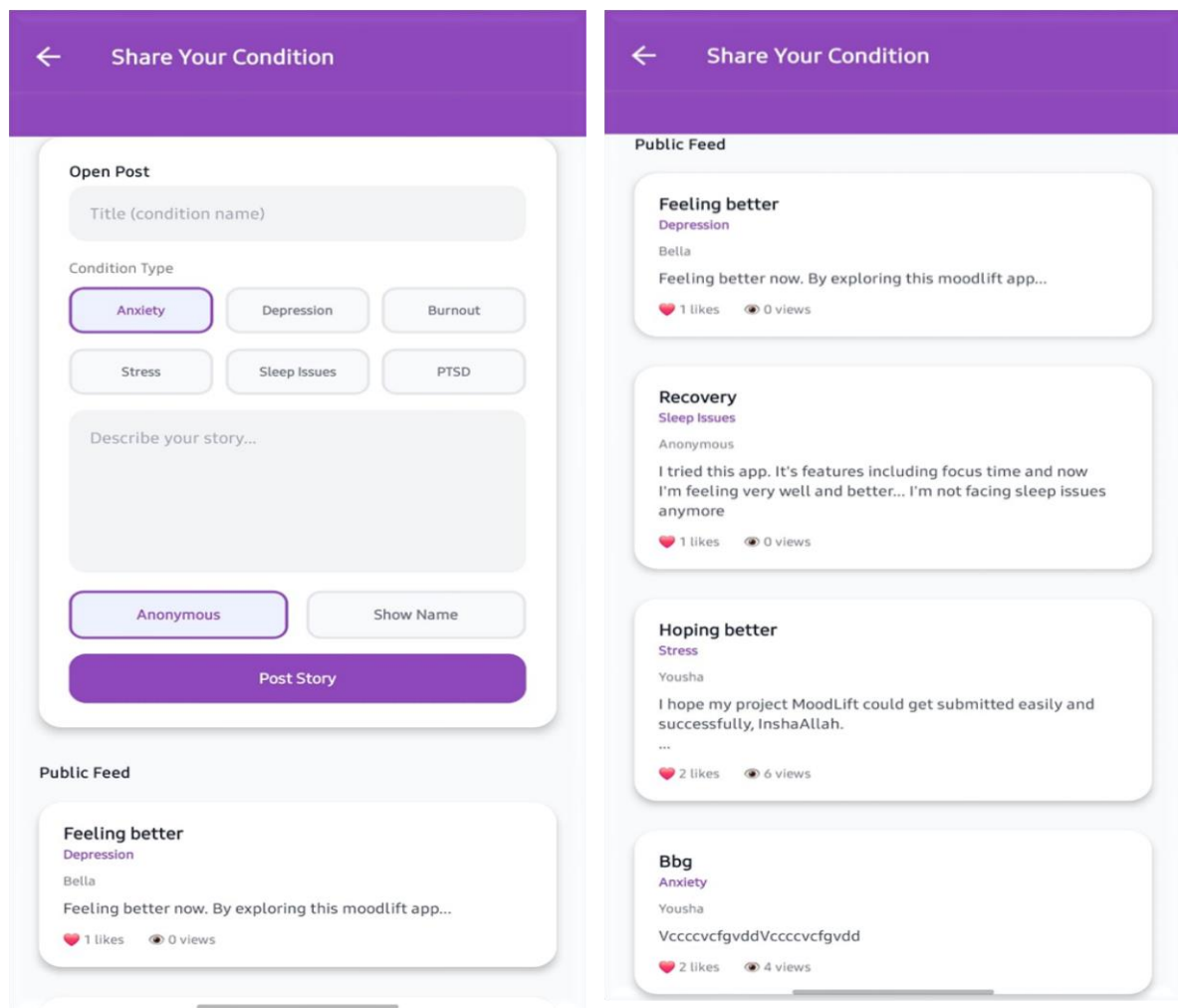


Figure 3.16: Community Interaction of MoodLift

Other MoodLift Screens Preview:

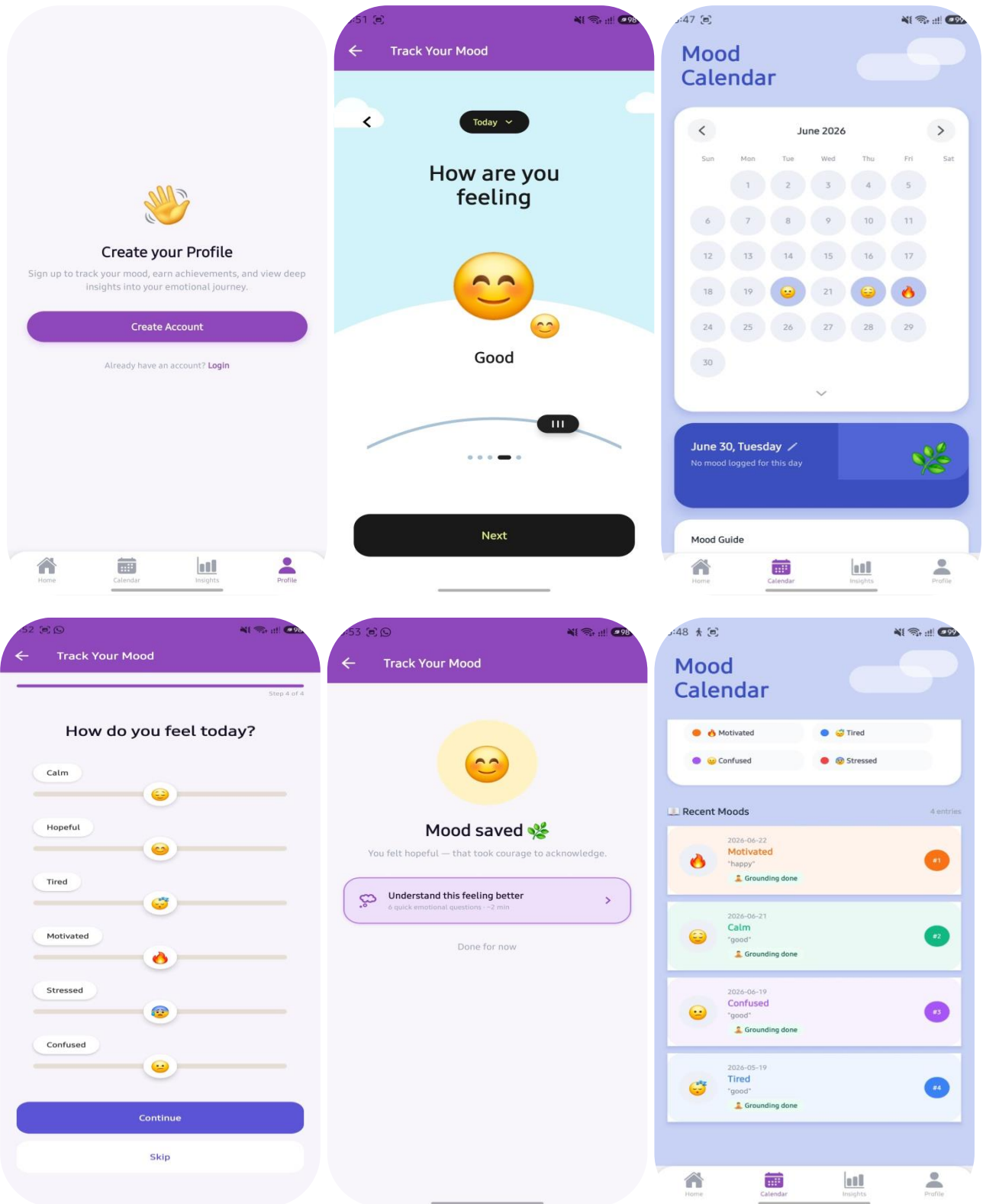


Figure 3.17: Other UI Screens

3.8 Backend Design of MoodLift

The Backend architecture of the application is act like its backbone, which processes user requests through logic, checks users and sends requests back to the mobile application. MoodLift uses Node.js and Express.js to develop APIs that allow the React Native frontend access to the MongoDB DB in a secure and efficient manner.

The MoodLift backend has routes, controllers, models and middleware for application's functions. The modularity of the design provides better readability and maintainability.

3.8.1. Backend Architecture

The back end architecture consists of many components that work together to process user actions.

When a user performs an action within the mobile application (for example, logs in, creates a mood, shares a story, or comments on someone's post), the request is sent to the Express server. The Express server processes the request, interacts with the MongoDB database through Mongoose, and sends the appropriate response back to the mobile application.

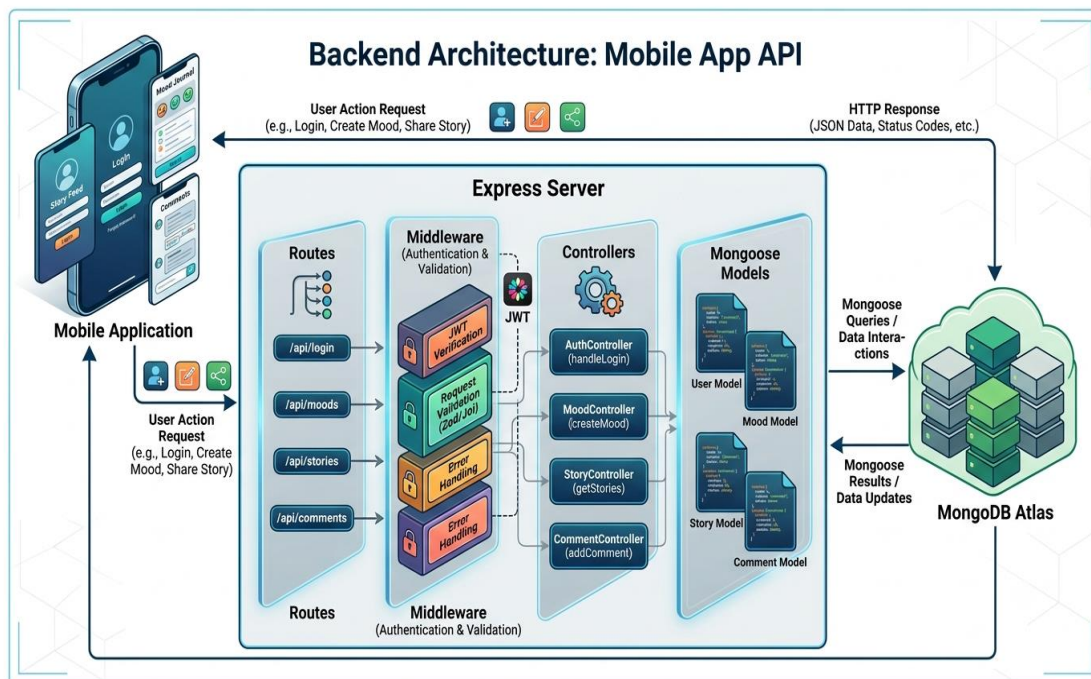


Figure 3.18: Backend Architecture

3.8.2 MoodLift's API Design

The MoodLift app communicates with the server using APIs. Each API has a specific purpose.

Table 3.3: MoodLift's API Design

API Endpoint	HTTP Method	Purpose
Register	POST	Register a new user
Login	POST	Authenticate User
Profile	GET	Retrieve user profile
Profile(Update)	PUT	Update profile information
Mood	POST	Saves daily mood
Stories	GET	Retrieve mood history
Stories (Comment)	POST	Add a comment
Stories (Like)	POST	Like or Unlike story
Questions	GET	Retrieve daily mood related questions
Logout	POST	Logout user

Data is transferred in the JSON format from the front end of the MoodLift application to the backend of the MoodLift server using the APIs, thus allowing an easy and efficient process for both types of systems to send and receive data between the frontend and backend of the application.

3.8.3 Authentication and Authorization in MoodLift

Securing the MoodLift application is a significant priority for mood lift and is therefore accomplished through both an authentication and authorization process to ensure that only users registered within the application can gain access to the protected resources.

Following steps are used in MoodLift to authenticate users.

1. The server verifies the user data.
2. The MoodLift server creates a JWT Token for the user.
3. The MoodLift server sends via the mobile app interface the JWT Token to the user.
4. The mobile app securely stores the JWT Token.
5. Every API call to a protected resource includes the JWT Token.
6. The backend validates the JWT Token on requests.

The user will also have their passwords encrypted prior to being saved in a MongoDB database using a bcrypt hash process to ensure that the password information will remain safe and secure even if there were to be a breach of the database.

Each of these security features in place by MoodLift will allow the user of the application to have security over their data while preventing unauthorized access

3.9 Database Design of MoodLift

MoodLift application utilized a NoSQL database called MongoDB for storing application's data. MongoDB stores all type of data in the structure of collections and documents which offers scalability and Flexibility which is suitable for dynamic and rapidly changing application data structures. The database consists of user's information, mood entry data, stories data, comments data, likes data, wellness question's data and data of the profile.

The usage of MongoDB enables greater speed and extensibility of MoodLift app without changing database design.

3.9.1 ER Diagram

ER Diagram represents the relationships between different entities used in the MoodLift system, such as user, mood, story, comment and likes. It helps in understanding how data is connected and managed. This diagram shows how users interact with different modules through mood tracking, story sharing, comments, likes, and notification activities.

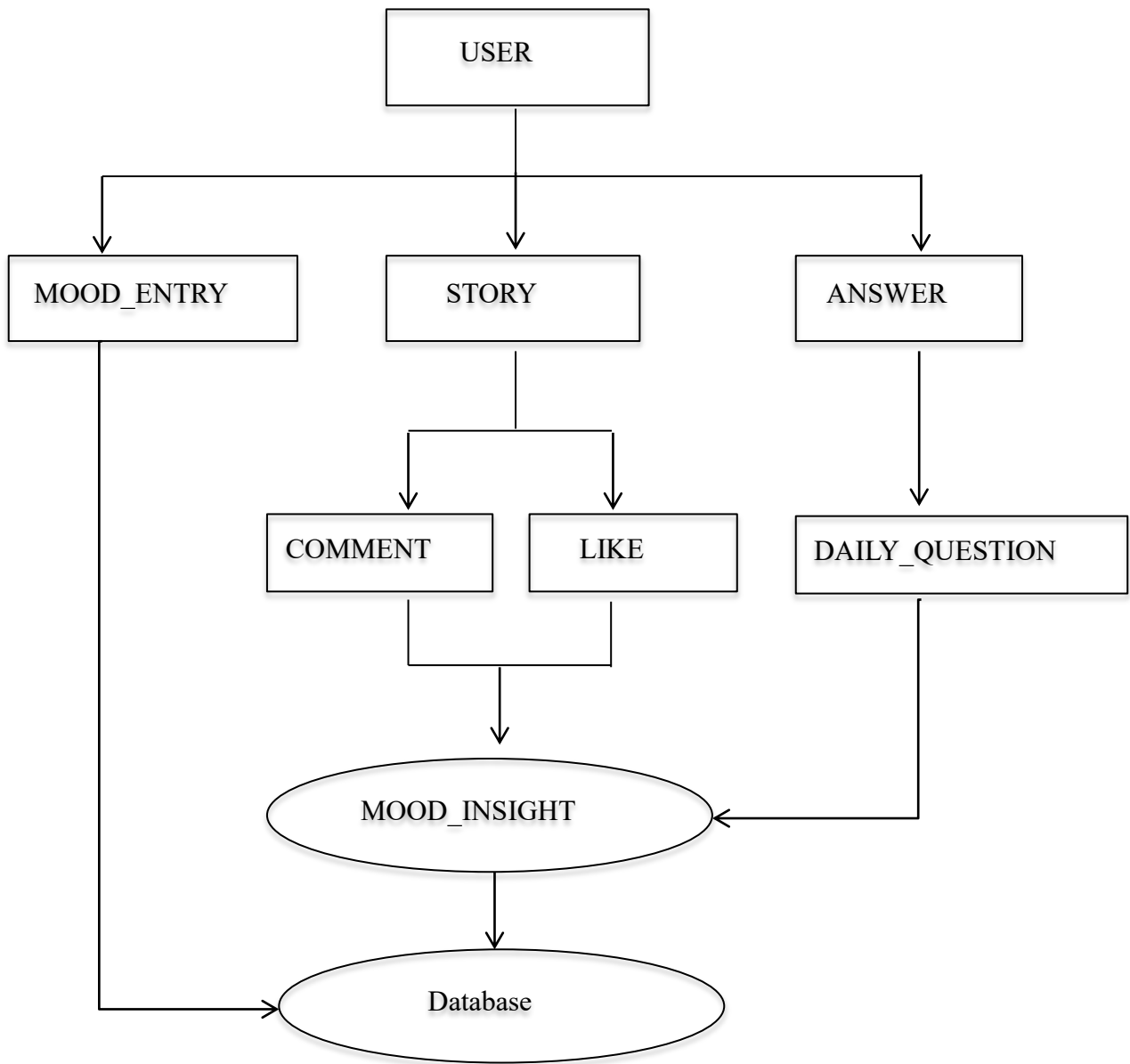


Figure 4.1: ER Diagram

Chapter 4: Testing and Evolution

4.1 Testing Strategy

Testing is an integral step in software development. The purpose of testing is to check whether the program is functioning as required result and whether it satisfies all requirements or not. The main focus of testing is to find and fix all bugs before releasing the application. A properly testing provides a higher level of reliability, security, and user satisfaction. The testing for the MoodLift mobile app was performed during all phases of the program individually. Each feature was tested independently and collectively with all other features within the application to guarantee that all functional features work seamlessly together. The main areas of testing are user authentication, mood tracking, community interaction, profile management, and backend API integration.

The following methods of testing were used during the development:

Functional Testing, Integration testing, User Interface testing, API testing, Database testing, and User Acceptance Testing (UAT). All these types of testing methods ensured the working of the application for normal operating conditions.

4.2 Test Cases

The purpose of creating test cases was to identify whether all of the main features of the MoodLift application function as they are intended. This is a list of the most important test cases executed during the evaluation phase.

Table 4.1: Test Cases

Test Case ID	Feature	Test Scenario	Expected Result	Status
TC-01	Registration	with valid data	Successfully created	Pass
TC-02	Registration	With invalid data	Not Created	Fail
TC-03	Login	With valid data	Login successfully	Pass
TC-04	Login	With invalid data	Error message alert	Fail

Test Case ID	Feature	Test Scenario	Expected Result	Status
TC-05	Mood Tracking	Save daily mood	Successfully created	Pass
TC-06	Daily questions	Submit response	Submitted successfully	Pass
TC-07	Share story	Publish story	Publish successfully	Pass
TC-08	Like story	Like a story	Story liked	Pass
TC-09	Profile	Update profile	Data updated	Pass
TC-10	Logout	Logout from app	User session end successfully	Pass

4.3 Testing Results

The results of the test determined that the MoodLift application provides a high level of consistency and dependability within all of its main functional areas. Each major feature of the application:

User authentication, mood tracking, wellness questions, making connections with other users in the MoodLift community, and profile management has been successfully and accurately tested. The application was able to successfully communicate with its back-end server using RESTful APIs and was able to store and retrieve all users from the MongoDB database. During the development phase of the application, minor imperfections were found in regards to the user interface's alignment and the providing of input validation messages. However, these issues were resolved prior to the final testing. The final evaluation and testing phase did not detect any major defects that would have negatively impacted the core functionality of the application.

Table 4.2: Summary of results

Category	Result
User Registration	Passed
User Login	Passed
Mood Tracking	Passed
Story Sharing	Passed
Personal Story	Failed
Like & Comment Story	Passed
Personal Contact	Failed
Profile Management	Passed
Database Operations	Passed
API Communication	Passed
Notification	Failed
AI Chatbot	Failed

Overall the testing result shows that MoodLift meets the requirements as specified or listed in the SRS of the project. The system proved to be stable, secure, and easy to use existing features while enabling users to monitor moods and access communities. And failed outcomes in the given table means these features are not yet included in the app that's why it is showing failed. It will surely added in the next phase or next update.

Chapter 5: MoodLift's Summary, Future Work and Conclusion

5.1 Project Summary

The project was concerned with building MoodLift, a cross-platform mobile application with the aim of improving mental well-being through providing tools to observe and record their emotional health. To this effect, we used React Native (Expo) for frontend, Node.js, Express.js, and MongoDB for the back-end, and the database respectively.

We integrated a few basic features which are: -

User authentication, Recording daily moods and taking wellness tests, Storytelling, Posting experiences from individual journey, Interaction with Community Stories, Displaying mood observations of individual on graphs and trends and many more.

The development followed a life cycle that includes but not limited to;

Requirement Analysis, System Design and Modeling, Database Design, API modeling, Testing and evaluation, Deployment and maintenance.

MoodLift had easy-to use-interface and requires no technical knowledge to operate by user. With this project, we established the potential of mobile applications in fostering mental health and emotional stability.

5.2 Future Enhancements

Despite having accomplished its current goal, below are few possible future improvements that can be added to the MoodLift application in the future

1. Integrating a chatbot using artificial intelligence which can assist user on basic emotional support and guidelines.
2. Recommending health and wellness tips personalized by mood trends of the user.
3. Incorporating push notifications which alert the user to enter there mood for the day.
4. Adding emergency contact person and integrating a national mental health hotline into the app.
5. Adding a mode for user voice journal entry instead of written text, this may also be

referred to as the ‘verbal feedback feature.’

6. Adding a mode that can allow for the prediction of your next emotional state based on advance algorithm machine learning.

7. Adding detailed reports and charts about mood patterns over time which may be called ‘The Happiness Tracker’.

8. Providing private user groups for the users who may want to socialize and talk to users that have similar feelings and interests.

9. Adding support for many languages that may be used internationally to access the application, which will be called ‘Global Support.

10. Deploying on Both Android’ and ‘iOS app stores would lead in increased and high user engagement.

5.3 Conclusion of our Project

In today’s world, mental health is a very relevant and significant issue and it is a must to keep a check over one’s emotional stability as well. Being able to monitor your feelings and reflecting about what you feel about is very vital. For that, a mobile application named MoodLift has been designed, it helps you keep your mood daily, makes you think about your feelings, interact with your buddies or the community to feel better.

The **MoodLift** project fulfilled its requirements by successfully creating a mobile application that offers functionality in term of mood tracking, wellness questions, sharing story and mood insights. It works seamlessly, providing a safe and very interactive interface and by making the application more engaging. MERN stack with react-native has helped me build a scalable, user friendly with amazing front end as well as backend,

As far as testing is concerned, through the various tests executed on the app, it performed to its expectations and will fulfill the need of what was requested. Therefore I can say, MoodLift may help the users to take a good care of their mental health. In future, with some new features this application can prove to be more beneficial in enhancing the emotional health of the users.